Ludwing Valdés Reyes

Software Engineer · Backend & Enterprise Systems

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PROFILE

Software engineer focused on backend, integrations and business systems. I turn manual processes into software that plant and office teams use every day, connected to the ERP, CAD and databases a company already runs. My CAD-to-ERP integration cut the time to load an assembly from 9 minutes to under 1.5.

EXPERIENCE

VLUX · Co-founder · Technology & product (personal project) — Aug 2026 - Present

- Lead technology for four business product lines: CRM, point of sale, industrial costing and quality management.
- Design and build integrations with ERPs, CAD/PDM, legacy databases, messaging webhooks and devices.

Industrial manufacturing company · IT & Systems — Sep 2024 - Aug 2026

- Automated BOM loading from CAD to the ERP: 9:08 → 1:28 min per assembly on 6 workstations, every write verified.
- Built the attendance and overtime system that takes biometric clocks to SQL Server and delivers HR reports in their own template.
- Owned support, networks, servers, backups and equipment maintenance in an ISO 9001 operation.

TECHNICAL SKILLS

Languages	Python, C# / .NET 8, TypeScript, SQL
Applied AI	AI to speed up real systems: automating flows, analyzing data and accelerating development
Backend & cloud	Node.js, REST APIs, webhooks, Supabase, Cloudflare, Vercel deployments
Data	SQL Server, PostgreSQL, Pervasive PSQL, Prisma; Excel data cleanup and material standardization
ERP & CAD	Global Shop, Odoo Community (custom add-ons); SolidWorks and AutoCAD APIs to generate BOMs
Frontend & desktop	React, Next.js, Electron, WinForms, PySide6 / Qt, PWA
Infra & quality	Networks, servers, backups, Git; pytest, Vitest, Playwright; ISO 9001 support

SELECTED PROJECTS

CAD-to-ERP integration · Python · C# / .NET 8 · SolidWorks API · SQL Server — In production

GOAL	Remove the manual, error-prone BOM entry between engineering and the ERP.
MY PART	Designed and built the whole app, from reading SolidWorks/PDM to validated writes in the ERP and change emails.
RESULT	About 6× faster end to end (9:08 → 1:28), backed by 1,737 automated tests.

Commercial CRM · Next.js · TypeScript · Prisma · PostgreSQL — In operation

GOAL	One traceable flow from lead to quote, project and warranty for a B2B installer.
MY PART	Architecture and full-stack build: guided quoting, price catalog, idempotent web intake.
RESULT	Web leads arrive on their own without duplicates, and no quote leaves without human review.

Attendance & overtime · C# · .NET 8 · SQL Server · ZKTeco SDK — In operation

GOAL	Stop HR from rebuilding clock data by hand every period.
MY PART	Layered .NET 8 solution: clock sync to SQL Server and an overtime rules engine.
RESULT	HR gets period reports in Excel and PDF, in their own template.

Also: ISO 9001 quality management system (demo), real-estate operations prototype and a point of sale on Odoo (pilot candidate). Full breakdown in the portfolio.